

Yingrui Hou

Data Scientist | Exploratory Modelling, Machine Learning & Validation

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PROFESSIONAL SUMMARY

PhD-trained data scientist with four years of postdoctoral research at CNRS, developing Python/C++ analytical workflows, machine learning models and statistical validation methods. Experienced in turning open research questions into reproducible studies, evaluating methods under noisy measurements and distribution shifts, and communicating assumptions and limitations to international collaborators. Interested in exploratory data and AI projects connecting technical feasibility with practical decisions.

CORE SKILLS

- Scientific programming: Python, NumPy, pandas, scikit-learn, statsmodels; C++, ROOT/RooFit, Linux and Git.
- Machine learning: supervised classification, feature studies, train/test diagnostics, threshold selection, sample reweighting and validation under distribution shift.
- Statistical modelling: likelihood estimation, mixture models, time-dependent inference, joint fitting, uncertainty quantification and sensitivity studies.
- Exploration and validation: proof-of-principle studies, control samples, calibration, monitoring, anomaly diagnosis and comparison of modelling assumptions.
- Technical delivery: reusable analysis components, reproducible configurations, collaborative review, documented findings and scientific presentations.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS - LPCA, Clermont-Ferrand | 2023-2025

- Developed a measurement-to-simulation workflow for GRAiNITA, translating prototype observations into response maps and Geant4 performance studies.
- Built calibration and signal-extraction methods using noise correction, channel equalisation and statistical fitting to produce comparable, position-resolved measurements.
- Integrated measured response maps into a simplified full-size model to estimate performance limits, including a non-uniformity constant term below 1% in the studied configuration.
- Developed time-profile fitting and correction methods for scintillation simulations, improving energy resolution by about a factor of 2 in a representative study.
- Compared material-dependent signal separation and communicated model assumptions, limitations and results to guide collaborative R&D discussions.

Postdoctoral Researcher | CNRS - LAPP, Annecy | 2021-2023

- Developed reusable Python/C++ workflows for weak-signal extraction and time-dependent inference on noisy datasets without direct event-level ground truth.
- Built machine learning classifiers with feature studies, train/test diagnostics and threshold scans to assess overfitting and validate event selection.
- Aligned control and simulation samples through machine learning reweighting to address distribution differences before downstream inference.
- Built monitoring and reconstruction checks that traced discrepancies to calibration, sensor or software causes, including a missing-energy software issue subsequently corrected downstream.

SELECTED TECHNICAL PROJECTS

Proof-of-principle validation | Advanced statistical inference, LHCb

- Contributed to testing a novel unbinned inference method under background, efficiency and resolution effects, using pseudoexperiments and covariance-aware checks.
- Prepared real-data inputs through reconstruction, selection and statistical background subtraction to assess whether the method remained usable beyond idealised conditions.

Machine learning validation | LHCb

- Combined feature assessment, score validation, threshold scans and sample alignment into reproducible workflows to evaluate model reliability under dataset mismatch.

Latent-parameter estimation | Time-dependent and weak-signal analyses

- Modelled acceptance and resolution with dataset-specific corrections and joint fits to infer latent behaviour while accounting for measurement bias.
- Implemented signal-plus-background likelihood fits and automated yield, width and stability checks across datasets and selections.

Sensor modelling and signal decomposition | GRAiNITA and PSD studies

- Extracted spatial response maps from a 16-channel CERN test-beam prototype to constrain a physics-based model with measured inputs.
- Compared ZnWO₄ and BGO simulations from 1 to 20 GeV, using fitted timing components to assess reconstruction performance and limits of signal separation.

Reusable scientific software | QCFilter / STCF studies

- Developed a C++ package integrated into Gaudi, converting measured constraints into stochastic event filtering to restore missing correlations in a fast simulation.

EARLIER EXPERIENCE

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed analyses combining simulation, statistical modelling and calibration to validate models against experimental data.

EDUCATION

Joint Master and PhD in Particle Physics | University of Chinese Academy of Sciences | 2016-2021

BSc in Applied Physics | China University of Mining and Technology | 2010-2014

LANGUAGES

English: C1 | French: B1 progressing towards B2 | Chinese: native